

Module 03: Disk Scheduling — FCFS and SSTF

1. Algorithm Principles

- **FCFS:** Pure FIFO order. Fair with no starvation, but causes wild head swings.
- **SSTF:** Greedy local minimum search. Drastically reduces seek time, but causes **Starvation**.

Disk Scheduling: FCFS vs SSTF Head Movement Comparison

Queue: 98, 183, 37, 122, 14, 124, 65, 67 | Initial Head Position = 53

1. FCFS (First-Come, First-Served)

Sequence: 53 → 98 → 183 → 37 → 122 → 14 → 124 → 65 → 67

Calculations:

- $|98 - 53| = 45$
- $|183 - 98| = 85$
- $|37 - 183| = 146$ (Wild swing across disk!)
- $|122 - 37| = 85$
- $|14 - 122| = 108$
- $|124 - 14| = 110$
- $|65 - 124| = 59$
- $|67 - 65| = 2$

Total Head Movement = 640 Cylinders

- **Drawback:** Zero optimization; wild head swings.
- **Advantage:** Simple, fair, no starvation.

2. SSTF (Shortest Seek Time First)

Sequence: 53 → 65 → 67 → 37 → 14 → 98 → 122 → 124 → 183

Calculations (Always pick nearest cylinder):

- $|65 - 53| = 12$
- $|67 - 65| = 2$
- $|37 - 67| = 30$
- $|14 - 37| = 23$
- $|98 - 14| = 84$
- $|122 - 98| = 24$
- $|124 - 122| = 2$
- $|183 - 124| = 59$

Total Head Movement = 236 Cylinders (63% Less!)

- **Serious Flaw:** Starvation of far requests!
- Greedy algorithm; not globally optimal.

2. Solved Numerical Summary

Queue: 98, 183, 37, 122, 14, 124, 65, 67 | Head = 53

- **FCFS Head Movement: 640 Cylinders**
- **SSTF Head Movement: 236 Cylinders** (63% reduction!)